

ChE 210 – Material and Energy Balance –Fall 2023

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #1)

SLO 1 Performance Indicator (PI)	(3) Satisfactory	(2) Developing	(1) Unsatisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject	Able to successfully combines mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development	Does not understand the connection between mathematical models and the system or process to be analyzed or designed
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra	applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems
4. Translates academic theory into engineering applications	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality	Some gaps in understanding the application of theory to the problem and expects theory to predict reality	Does not appear to grasp the connection between theory and the problem

Averages:

Performance Indicators: (1) 2.40 (2) 1.93 (4) 1.93 Overall Average: 2.09

PI1 is the strongest score by a clear margin. This score was evaluated on students' ability to carry out numerical calculations for material balance equations for a multicomponent, multiunit operations process with reaction. The calculations were made during the final exam using pocket calculators. The students who struggle with this task used calculations that were not based on concepts of mass conservation, or made numerical errors using their calculators. Some students do fail to understand that ad hoc combinations of values appearing in the problem statement will not satisfy conservation of mass. Although this is a central principle of the course, it can be re-enforced by more specific exercises in the discussion portion of the course

The PI2 is tied with PI4 at an average of 1.93. Here we focus on the solution of the material balance equations after they are set up. This usually requires algebra for steady state problems and differential equations for transient or unsteady equations. Once the student has developed the material balance equations correctly, can the student correctly solve them? Students arrive in ChE 210 with inadequate math skills, even at the level a solving systems of algebraic equations. Solving linear and non-linear algebraic equations is a prerequisite skill for the course; however, students' proficiency has decreased recently. Most students will have taken CHE 205 which is a computer applications course that covers solving systems of linear and non-linear algebraic equations.

Finally, PI4 which corresponds to being able to set up the material balance equations for the chosen systems. For this example, students needed to define systems around unit operations and groups of unit operation. For a

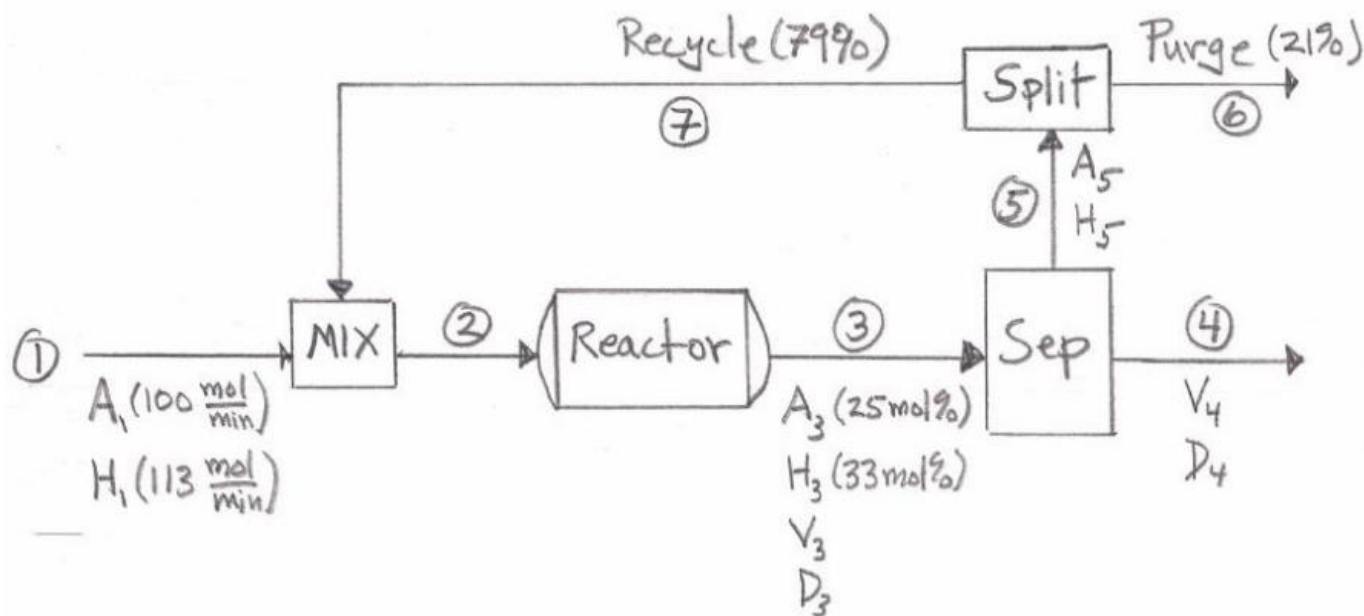
system, they must identify the input and output stream variable for the four species being processed. There is some mixed success for students doing this. It is perhaps the most difficult new skill they need to acquire, that is, engineering analysis. Students need to recognize whether a system is well posed or under specified. This problem is completely linear and can be solved with linear equations only. The students who had trouble with the problem struggled with setting up a set of solvable equations. In the future, more examples of this type will be added to the discussion section for practice.

1. (100 points) A gas-phase process is designed to react acetylene (C_2H_2) and hydrogen chloride (HCl) to produce vinyl chloride (C_2H_3Cl). An undesired reaction further reacts the vinyl chloride to dichloroethane ($C_2H_4Cl_2$). The two reactions taking place in the reactor are:



The reactions are written twice. The set of second reaction equations are written in a useful shorthand where A=Acetylene, H=Hydrogen chloride, V=Vinyl chloride, and D=Dichloroethane and is used in the block flow diagram below. There are seven streams labeled from 1 to 7. Use the single-letter shorthand with the stream number as a subscript to indicate the species molar flow rate for that stream.

A fresh feed (Stream 1) of 100 molA/min and 113 molH/min enters a mixer. A recycle stream containing A and H (Stream 7) returning from the separator also enter the mixer. The exiting stream from the mixer (Stream 2) is fed into a reactor. The output of the reactor (Stream 3) is 25mol%A and 33mol%H. This stream is sent to a separator that has a product stream of V and D (Stream 4). Also, a second stream exits the separator (Stream 5) which goes to a splitter that divides the stream into a purge stream (Stream 6) and the recycle stream (Stream 7) back to the mixer. The purge (Stream 6) is 21% of Stream 5.



- (a) (15 points) Do a DOF analysis for the entire process to show that the problem is well posed. Follow the method on page 105 of Murphy. Fill in the table below and provide a brief but clear explanation for the values you assigned for each row. You must justify the value you enter or no credit will be given!

	Value	Briefly explain the value you give
4a		
4b		
Independent Variables		$4a + 4b$
5a		
5b		
5c		
5d		
Independent Equations		$5a + 5b + 5c + 5d$
DOF Total		$(4a + 4b) - (5a + 5b + 5c + 5d)$

- (b) (85 points) Calculate all of the molar flow rates in mol/min for the process and fill in the table below. Feel free to re-draw the table yourself. You must clearly show your calculations for these values to receive full credit. Use at least three significant figures for your calculations.

Streams ->	1	2	3	4	5	6	7
A (mol/min)	100			-----			
H (mol/min)	113			-----			
V (mol/min)	-----	-----			-----	-----	-----
D (mol/min)	-----	-----			-----	-----	-----
Total $\dot{n}$ (mol/min)	213						